

Miniproject 2 - Creating a ZCB Term Structure

FRE 6103 Valuation for Financial Engineering, NYU Tandon | Raj Pawar | WSJ Treasury quotes of Sep 4, 2026, settlement Sep 8, 2026

Data and conventions. Wall Street Journal (Tullett Prebon) 3 pm quotes for all 353 US Treasury notes and bonds on Friday, September 4, 2026, the Week 2 data file. Quotes are clean prices in 32nds with a third decimal in eighths of a 32nd ($99.256 = 99 + 25.75/32$); we price off the asked side. Regular-way settlement is T+1 business days, Tuesday, September 8, 2026 (Monday is Labor Day). Both conventions were checked by recomputing every WSJ asked yield: median discrepancy 0.02 bp and RMSE 0.5 bp, against 1.8 bp for the next-best settlement date and over 86 bp if the quotes are read as decimals.

Cash flows. Each bond pays coupon/2 semi-annually on its maturity day-of-month (month-end maturities pay on month-ends) and 100 at maturity. Accrued interest is actual/actual within the coupon period, coupon/2 $\times$ days accrued / days in period; dirty price = clean price + accrued (slide 34). Time to each payment is $t = \text{days}/365$ from settlement. Street yields follow the Treasury convention (semi-annual compounding with a fractional first period, simple interest when one payment remains), which reproduces Excel's YIELD.

Model. The continuously compounded zero rate follows the Svensson (1994) form suggested with the data file, the discount factor is $d(t) = e^{-r(t)t}$, and each bond's model dirty price is the sumproduct of its payment stream with the discount factors (slide 33):

$$r(t) = \beta_0 + \beta_1 \frac{1 - e^{-t/\tau_1}}{t/\tau_1} + \beta_2 \left(\frac{1 - e^{-t/\tau_1}}{t/\tau_1} - e^{-t/\tau_1} \right) + \beta_3 \left(\frac{1 - e^{-t/\tau_2}}{t/\tau_2} - e^{-t/\tau_2} \right)$$

$$d(t) = e^{-r(t)t}, \quad P_i^{\text{model}} = \sum_k CF_{ik} d(t_{ik})$$

The form nests Nelson-Siegel ($\beta_3 = 0$), is smooth, has finite limits at both ends ($r(0) = \beta_0 + \beta_1$, $r(\infty) = \beta_0$) and gives the instantaneous forward curve in closed form, $f(t) = \beta_0 + \beta_1 e^{-t/\tau_1} + \beta_2 (t/\tau_1) e^{-t/\tau_1} + \beta_3 (t/\tau_2) e^{-t/\tau_2}$.

Estimation. The six parameters minimise the sum of squared price errors, each divided by dirty price times modified duration D_i , which is the first-order yield error; the objective is therefore the sum of squared yield errors (slide 31) while every model price stays a linear sumproduct:

$$\min_{\beta, \tau} \sum_{i \in \text{fit}} \left(\frac{P_i^{\text{model}} - P_i^{\text{market}}}{P_i^{\text{market}} D_i} \right)^2, \quad \beta_0 \geq 0, \tau_1, \tau_2 \geq 0.05$$

Bonds with under 3 months to maturity (14 of 353) are excluded from estimation, following Gürkaynak, Sack and Wright (2007): their prices reflect money-market conditions and, with near-zero duration, a one-cent quote error is a 50 bp yield error; they are still priced off the curve as an out-of-sample check. The objective is multimodal in (τ_1, τ_2) , so the least-squares solver is started from 19 points on a τ grid; 11 reach the same optimum. In Excel the same objective is the sum-of-squares cell on the Curve sheet, minimised with Solver (GRG Nonlinear); the workbook reproduces every Python model price, yield and zero rate.

Results. $\beta_0 = 0.0000$ (bound active), $\beta_1 = 0.0363$, $\beta_2 = 0.0447$, $\beta_3 = 0.1622$, $\tau_1 = 1.76$, $\tau_2 = 15.93$ (Table 1). The curve prices the 339 bonds with a yield RMSE of 2.9 bp (mean absolute 2.1 bp; worst 14.8 bp, the 4.375% Dec 15, 2026 note); Nelson-Siegel gives 7.5 bp, so the second hump is needed. The zero curve rises from 3.63% at the short end to 4.14% at 1 year, 4.49% at 5, 4.79% at 10, 5.33% at 20 and 5.31% at 30 years, peaking at 5.38% near 24 years (Figure 1, Table 2). Table 3 lists $r(t)$, $d(t)$ and $f(t)$ for all 245 distinct Treasury payment dates from Sep 15, 2026 to Aug 15, 2056. β_0 , the rate at infinite maturity, is weakly identified by a 30-year sample: pinning it anywhere from 0% to 3% worsens the fit by only 0.3 bp RMSE and moves the in-sample curve by about 10 bp at the two ends (3% in $d(30y)$), so the payment-date discount factors are robust to it; with $\beta_0 = 0$ the forward rate decays beyond the sample, so the curve should not be extrapolated past 30 years.

Results

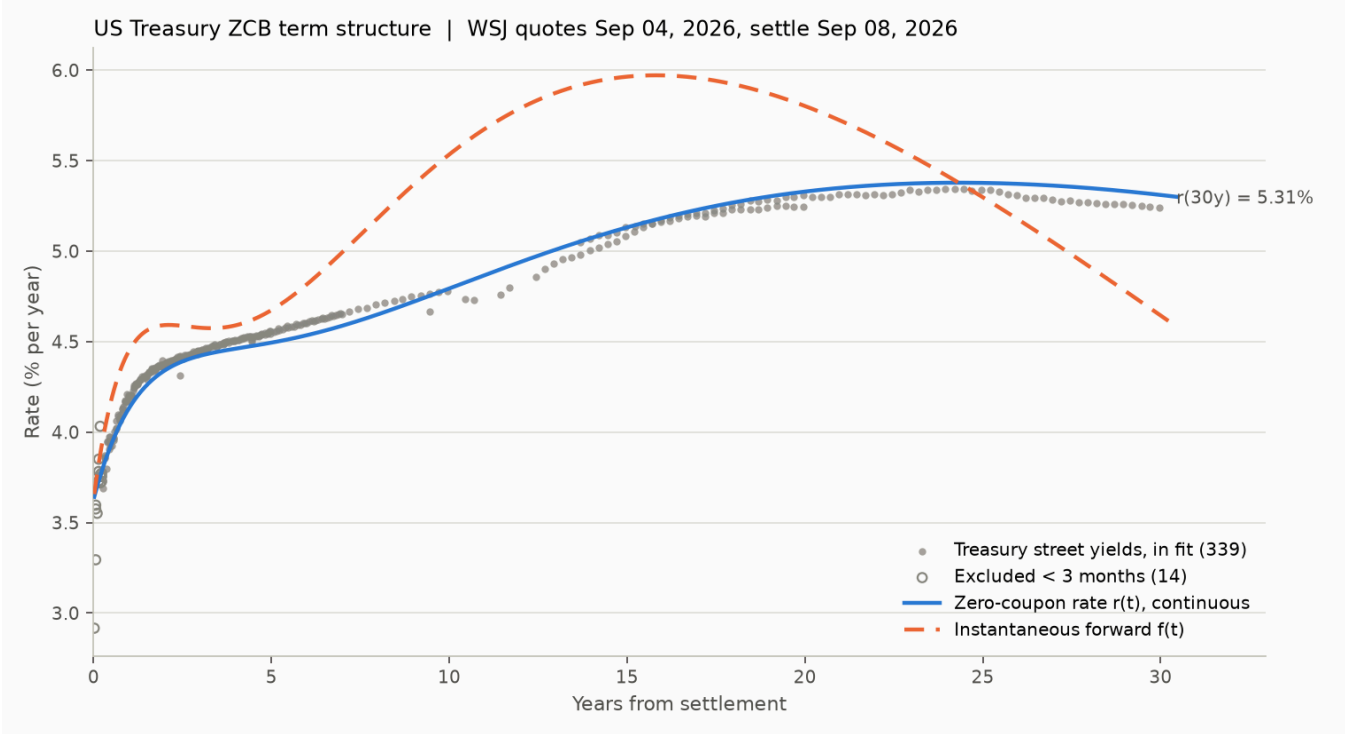


Figure 1. Fitted continuously compounded zero curve $r(t)$ and instantaneous forward curve $f(t)$ against the street yields of the 339 bonds in the fit (dots) and the 14 excluded bonds under 3 months (hollow circles). Sep 4, 2026 quotes, settlement Sep 8, 2026.

Table 1. Parameters and fit quality (bonds in fit).

	Svensson	Nelson-Siegel
β_0	0.00000	0.05879
β_1	0.03628	-0.01870
β_2	0.04467	-0.00000
β_3	0.16218	0.00000
τ_1 (years)	1.761	7.015
τ_2 (years)	15.927	—
Bonds in fit	339	339
Objective (Σ squared yield errors)	2.810e-05	1.922e-04
Yield RMSE (bp)	2.88	7.52
Yield mean absolute error (bp)	2.06	5.85
Largest yield error (bp)	14.8	38.9
Price RMSE (per 100 face)	0.185	0.382
Starting points reaching optimum	11 / 19	4 / 4

Table 2. Zero curve at selected maturities, continuously compounded.

t (years)	r(t) %	d(t)	f(t) %
0.25	3.796	0.99055	3.948
0.5	3.933	0.98053	4.179
1	4.133	0.95951	4.450
2	4.340	0.91685	4.591
3	4.422	0.87575	4.577
5	4.494	0.79875	4.674
7	4.588	0.72530	4.995
10	4.794	0.61918	5.534
15	5.131	0.46315	5.964
20	5.329	0.34445	5.802
25	5.377	0.26073	5.298
30	5.310	0.20329	4.645

All bonds and the full payment-date table are also in the Excel workbook (sheets Bonds and Curve) and in the repository's CSV files.

Table 3. Discount rate for every Treasury payment date

Union of all coupon and principal dates of the 353 bonds: 245 dates. t in years (actual/365) from settlement Sep 8, 2026; $r(t)$ continuously compounded zero rate; $d(t) = e^{-r(t)t}$; $f(t)$ instantaneous forward rate.

Payment date	t	r(t) %	d(t)	f(t) %	Payment date	t	r(t) %	d(t)	f(t) %	Payment date	t	r(t) %	d(t)	f(t) %
Sep 15, 2026	0.019	3.6419	0.99930	3.656	Aug 15, 2028	1.937	4.3323	0.91951	4.590	Dec 31, 2030	4.315	4.4713	0.82453	4.609
Sep 30, 2026	0.060	3.6716	0.99779	3.714	Aug 31, 2028	1.981	4.3380	0.91766	4.591	Jan 31, 2031	4.400	4.4740	0.82131	4.615
Oct 15, 2026	0.101	3.7004	0.99626	3.770	Sep 15, 2028	2.022	4.3432	0.91593	4.592	Feb 15, 2031	4.441	4.4753	0.81975	4.618
Oct 31, 2026	0.145	3.7299	0.99460	3.826	Sep 30, 2028	2.063	4.3481	0.91420	4.592	Feb 28, 2031	4.477	4.4765	0.81840	4.621
Nov 15, 2026	0.186	3.7567	0.99303	3.876	Oct 15, 2028	2.104	4.3529	0.91248	4.592	Mar 31, 2031	4.562	4.4792	0.81520	4.629
Nov 30, 2026	0.227	3.7826	0.99144	3.923	Oct 31, 2028	2.148	4.3578	0.91064	4.592	Apr 30, 2031	4.644	4.4820	0.81210	4.636
Dec 15, 2026	0.268	3.8076	0.98983	3.968	Nov 15, 2028	2.189	4.3622	0.90893	4.592	May 15, 2031	4.685	4.4833	0.81055	4.640
Dec 31, 2026	0.312	3.8334	0.98810	4.013	Nov 30, 2028	2.230	4.3664	0.90721	4.592	May 31, 2031	4.729	4.4848	0.80890	4.644
Jan 15, 2027	0.353	3.8567	0.98646	4.053	Dec 15, 2028	2.271	4.3705	0.90550	4.591	Jun 30, 2031	4.811	4.4876	0.80582	4.653
Jan 31, 2027	0.397	3.8806	0.98470	4.094	Dec 31, 2028	2.315	4.3746	0.90368	4.590	Jul 31, 2031	4.896	4.4905	0.80264	4.662
Feb 15, 2027	0.438	3.9023	0.98304	4.129	Jan 15, 2029	2.356	4.3784	0.90198	4.590	Aug 15, 2031	4.937	4.4920	0.80110	4.667
Feb 28, 2027	0.474	3.9205	0.98159	4.159	Jan 31, 2029	2.400	4.3822	0.90017	4.589	Aug 31, 2031	4.981	4.4935	0.79946	4.672
Mar 15, 2027	0.515	3.9408	0.97991	4.191	Feb 15, 2029	2.441	4.3857	0.89847	4.588	Sep 30, 2031	5.063	4.4965	0.79640	4.681
Mar 31, 2027	0.559	3.9617	0.97810	4.223	Feb 28, 2029	2.477	4.3886	0.89701	4.587	Oct 31, 2031	5.148	4.4996	0.79323	4.692
Apr 15, 2027	0.600	3.9805	0.97640	4.251	Mar 15, 2029	2.518	4.3919	0.89532	4.587	Nov 15, 2031	5.189	4.5012	0.79170	4.697
Apr 30, 2027	0.641	3.9987	0.97469	4.278	Mar 31, 2029	2.562	4.3952	0.89352	4.586	Nov 30, 2031	5.230	4.5028	0.79018	4.702
May 15, 2027	0.682	4.0163	0.97297	4.303	Apr 15, 2029	2.603	4.3982	0.89184	4.585	Dec 31, 2031	5.315	4.5060	0.78702	4.714
May 31, 2027	0.726	4.0344	0.97113	4.328	Apr 30, 2029	2.644	4.4011	0.89016	4.584	Jan 31, 2032	5.400	4.5094	0.78387	4.725
Jun 15, 2027	0.767	4.0507	0.96940	4.350	May 15, 2029	2.685	4.4039	0.88848	4.583	Feb 15, 2032	5.441	4.5110	0.78235	4.731
Jun 30, 2027	0.808	4.0664	0.96767	4.370	May 31, 2029	2.729	4.4067	0.88670	4.582	Feb 29, 2032	5.479	4.5126	0.78093	4.736
Jul 15, 2027	0.849	4.0816	0.96593	4.390	Jun 15, 2029	2.770	4.4093	0.88503	4.581	Mar 31, 2032	5.564	4.5161	0.77779	4.749
Jul 31, 2027	0.893	4.0972	0.96407	4.409	Jun 30, 2029	2.811	4.4118	0.88337	4.580	Apr 30, 2032	5.647	4.5196	0.77476	4.761
Aug 15, 2027	0.934	4.1113	0.96232	4.426	Jul 15, 2029	2.852	4.4142	0.88171	4.579	May 15, 2032	5.688	4.5213	0.77324	4.767
Aug 31, 2027	0.978	4.1258	0.96045	4.442	Jul 31, 2029	2.896	4.4167	0.87994	4.578	May 31, 2032	5.732	4.5233	0.77163	4.774
Sep 15, 2027	1.019	4.1388	0.95870	4.457	Aug 15, 2029	2.937	4.4190	0.87828	4.578	Jun 30, 2032	5.814	4.5269	0.76860	4.787
Sep 30, 2027	1.060	4.1514	0.95694	4.470	Aug 31, 2029	2.981	4.4213	0.87652	4.577	Jul 31, 2032	5.899	4.5307	0.76548	4.800
Oct 15, 2027	1.101	4.1635	0.95518	4.482	Sep 30, 2029	3.063	4.4255	0.87323	4.576	Aug 15, 2032	5.940	4.5326	0.76397	4.807
Oct 31, 2027	1.145	4.1760	0.95330	4.495	Oct 31, 2029	3.148	4.4295	0.86985	4.575	Aug 31, 2032	5.984	4.5347	0.76236	4.814
Nov 15, 2027	1.186	4.1872	0.95154	4.505	Nov 15, 2029	3.189	4.4314	0.86821	4.574	Sep 30, 2032	6.066	4.5385	0.75935	4.828
Nov 30, 2027	1.227	4.1980	0.94978	4.515	Nov 30, 2029	3.230	4.4332	0.86658	4.574	Oct 31, 2032	6.151	4.5426	0.75624	4.842
Dec 15, 2027	1.268	4.2084	0.94802	4.524	Dec 31, 2029	3.315	4.4368	0.86322	4.574	Nov 15, 2032	6.192	4.5446	0.75473	4.849
Dec 31, 2027	1.312	4.2191	0.94614	4.533	Jan 31, 2030	3.400	4.4403	0.85987	4.574	Nov 30, 2032	6.233	4.5467	0.75323	4.856
Jan 15, 2028	1.353	4.2287	0.94437	4.540	Feb 15, 2030	3.441	4.4419	0.85826	4.575	Dec 31, 2032	6.318	4.5509	0.75012	4.871
Jan 31, 2028	1.397	4.2386	0.94249	4.547	Feb 28, 2030	3.477	4.4432	0.85686	4.575	Jan 31, 2033	6.403	4.5553	0.74702	4.886
Feb 15, 2028	1.438	4.2475	0.94073	4.553	Mar 31, 2030	3.562	4.4464	0.85354	4.576	Feb 15, 2033	6.444	4.5574	0.74552	4.893
Feb 29, 2028	1.477	4.2556	0.93909	4.559	Apr 30, 2030	3.644	4.4493	0.85033	4.578	Feb 28, 2033	6.479	4.5593	0.74422	4.899
Mar 15, 2028	1.518	4.2638	0.93733	4.564	May 15, 2030	3.685	4.4508	0.84874	4.579	Mar 31, 2033	6.564	4.5638	0.74113	4.915
Mar 31, 2028	1.562	4.2723	0.93546	4.568	May 31, 2030	3.729	4.4523	0.84703	4.580	Apr 30, 2033	6.647	4.5682	0.73813	4.930
Apr 15, 2028	1.603	4.2800	0.93370	4.572	Jun 30, 2030	3.811	4.4550	0.84385	4.582	May 15, 2033	6.688	4.5704	0.73664	4.937
Apr 30, 2028	1.644	4.2873	0.93195	4.576	Jul 31, 2030	3.896	4.4579	0.84057	4.586	May 31, 2033	6.732	4.5729	0.73505	4.945
May 15, 2028	1.685	4.2944	0.93020	4.579	Aug 15, 2030	3.937	4.4592	0.83899	4.587	Jun 30, 2033	6.814	4.5774	0.73206	4.960
May 31, 2028	1.729	4.3017	0.92833	4.582	Aug 31, 2030	3.981	4.4606	0.83730	4.589	Jul 31, 2033	6.899	4.5822	0.72898	4.976
Jun 15, 2028	1.770	4.3082	0.92659	4.584	Sep 30, 2030	4.063	4.4633	0.83415	4.593	Aug 15, 2033	6.940	4.5846	0.72749	4.984
Jun 30, 2028	1.811	4.3145	0.92484	4.586	Oct 31, 2030	4.148	4.4660	0.83090	4.598	Aug 31, 2033	6.984	4.5871	0.72590	4.992
Jul 15, 2028	1.852	4.3205	0.92310	4.588	Nov 15, 2030	4.189	4.4673	0.82933	4.601	Nov 15, 2033	7.192	4.5994	0.71836	5.031
Jul 31, 2028	1.896	4.3267	0.92124	4.589	Nov 30, 2030	4.230	4.4686	0.82776	4.603	Nov 30, 2033	7.233	4.6019	0.71688	5.039

Table 3, continued (2 of 2).

Payment date	t	r(t) %	d(t)	f(t) %	Payment date	t	r(t) %	d(t)	f(t) %	Payment date	t	r(t) %	d(t)	f(t) %
Feb 15, 2034	7.444	4.6148	0.70927	5.079	Nov 15, 2041	15.197	5.1422	0.45773	5.967	Feb 15, 2053	26.458	5.3677	0.24167	5.116
May 15, 2034	7.688	4.6303	0.70050	5.125	Nov 30, 2041	15.238	5.1444	0.45661	5.968	May 15, 2053	26.701	5.3653	0.23869	5.085
May 31, 2034	7.732	4.6331	0.69893	5.134	Feb 15, 2042	15.449	5.1557	0.45090	5.970	Aug 15, 2053	26.953	5.3625	0.23566	5.053
Aug 15, 2034	7.940	4.6468	0.69147	5.173	May 15, 2042	15.693	5.1683	0.44438	5.971	Nov 15, 2053	27.205	5.3595	0.23268	5.020
Nov 15, 2034	8.192	4.6637	0.68247	5.221	Aug 15, 2042	15.945	5.1810	0.43774	5.971	Feb 15, 2054	27.458	5.3562	0.22977	4.987
Nov 30, 2034	8.233	4.6665	0.68100	5.228	Nov 15, 2042	16.197	5.1933	0.43120	5.970	May 15, 2054	27.701	5.3528	0.22700	4.955
Feb 15, 2035	8.444	4.6810	0.67351	5.268	Feb 15, 2043	16.449	5.2052	0.42476	5.967	Aug 15, 2054	27.953	5.3491	0.22419	4.921
May 15, 2035	8.688	4.6981	0.66487	5.312	May 15, 2043	16.693	5.2163	0.41863	5.963	Nov 15, 2054	28.205	5.3451	0.22144	4.887
May 31, 2035	8.732	4.7012	0.66333	5.320	Aug 15, 2043	16.945	5.2274	0.41239	5.958	Feb 15, 2055	28.458	5.3409	0.21874	4.854
Aug 15, 2035	8.940	4.7161	0.65599	5.357	Nov 15, 2043	17.197	5.2380	0.40625	5.951	May 15, 2055	28.701	5.3366	0.21617	4.821
Nov 15, 2035	9.192	4.7343	0.64716	5.402	Feb 15, 2044	17.449	5.2483	0.40020	5.943	Aug 15, 2055	28.953	5.3320	0.21357	4.787
Nov 30, 2035	9.233	4.7372	0.64572	5.409	May 15, 2044	17.696	5.2579	0.39438	5.934	Nov 15, 2055	29.205	5.3271	0.21102	4.753
Feb 15, 2036	9.444	4.7526	0.63837	5.444	Aug 15, 2044	17.948	5.2673	0.38853	5.924	Feb 15, 2056	29.458	5.3221	0.20851	4.719
May 15, 2036	9.690	4.7708	0.62983	5.485	Nov 15, 2044	18.200	5.2763	0.38278	5.913	May 15, 2056	29.704	5.3169	0.20611	4.685
May 31, 2036	9.734	4.7740	0.62832	5.492	Feb 15, 2045	18.452	5.2849	0.37712	5.900	Aug 15, 2056	29.956	5.3115	0.20370	4.650
Aug 15, 2036	9.942	4.7894	0.62115	5.525	May 15, 2045	18.696	5.2929	0.37174	5.887					
Nov 15, 2036	10.195	4.8081	0.61253	5.564	Aug 15, 2045	18.948	5.3007	0.36627	5.873					
Nov 30, 2036	10.236	4.8111	0.61113	5.570	Nov 15, 2045	19.200	5.3081	0.36090	5.857					
Feb 15, 2037	10.447	4.8267	0.60397	5.601	Feb 15, 2046	19.452	5.3151	0.35562	5.841					
May 15, 2037	10.690	4.8448	0.59575	5.635	May 15, 2046	19.696	5.3215	0.35060	5.824					
May 31, 2037	10.734	4.8480	0.59428	5.641	Aug 15, 2046	19.948	5.3278	0.34549	5.806					
Aug 15, 2037	10.942	4.8634	0.58733	5.668	Nov 15, 2046	20.200	5.3336	0.34048	5.787					
Nov 15, 2037	11.195	4.8819	0.57897	5.700	Feb 15, 2047	20.452	5.3391	0.33556	5.767					
Nov 30, 2037	11.236	4.8849	0.57762	5.705	May 15, 2047	20.696	5.3440	0.33088	5.747					
Feb 15, 2038	11.447	4.9002	0.57069	5.730	Aug 15, 2047	20.948	5.3487	0.32613	5.726					
May 15, 2038	11.690	4.9178	0.56275	5.757	Nov 15, 2047	21.200	5.3531	0.32147	5.704					
May 31, 2038	11.734	4.9210	0.56134	5.762	Feb 15, 2048	21.452	5.3571	0.31689	5.681					
Aug 15, 2038	11.942	4.9358	0.55463	5.784	May 15, 2048	21.699	5.3606	0.31249	5.658					
Nov 15, 2038	12.195	4.9536	0.54658	5.809	Aug 15, 2048	21.951	5.3639	0.30808	5.633					
Nov 30, 2038	12.236	4.9565	0.54528	5.812	Nov 15, 2048	22.203	5.3668	0.30374	5.609					
Feb 15, 2039	12.447	4.9711	0.53862	5.831	Feb 15, 2049	22.455	5.3694	0.29949	5.583					
May 15, 2039	12.690	4.9879	0.53101	5.852	May 15, 2049	22.699	5.3715	0.29545	5.558					
May 31, 2039	12.734	4.9909	0.52965	5.855	Aug 15, 2049	22.951	5.3734	0.29135	5.532					
Aug 15, 2039	12.942	5.0049	0.52322	5.871	Nov 15, 2049	23.203	5.3750	0.28732	5.505					
Nov 15, 2039	13.195	5.0216	0.51552	5.888	Feb 15, 2050	23.455	5.3762	0.28337	5.477					
Nov 30, 2039	13.236	5.0243	0.51428	5.891	May 15, 2050	23.699	5.3771	0.27962	5.450					
Feb 15, 2040	13.447	5.0380	0.50792	5.904	Aug 15, 2050	23.951	5.3778	0.27582	5.421					
May 15, 2040	13.693	5.0537	0.50057	5.918	Nov 15, 2050	24.203	5.3781	0.27209	5.392					
May 31, 2040	13.737	5.0565	0.49927	5.920	Feb 15, 2051	24.455	5.3781	0.26842	5.363					
Aug 15, 2040	13.945	5.0695	0.49315	5.930	May 15, 2051	24.699	5.3778	0.26494	5.334					
Nov 15, 2040	14.197	5.0848	0.48582	5.941	Aug 15, 2051	24.951	5.3772	0.26142	5.304					
Nov 30, 2040	14.238	5.0873	0.48464	5.943	Nov 15, 2051	25.203	5.3763	0.25795	5.273					
Feb 15, 2041	14.449	5.0999	0.47860	5.950	Feb 15, 2052	25.455	5.3751	0.25456	5.242					
May 15, 2041	14.693	5.1140	0.47170	5.957	May 15, 2052	25.701	5.3737	0.25130	5.212					
May 31, 2041	14.737	5.1165	0.47047	5.958	Aug 15, 2052	25.953	5.3720	0.24803	5.180					
Aug 15, 2041	14.945	5.1283	0.46467	5.963	Nov 15, 2052	26.205	5.3700	0.24482	5.149					